

BOWEN, AustraliaWMO: **943660**Lat: **20.0154S**Lon: **148.2140E**Elev: **26**StdP: **14.68**Time Zone: **10.00 (E10)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.9	52.3	34.0	28.7	62.6	38.4	34.2	62.8	27.5	71.1	25.1	70.9	6.7	230	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	90.7	78.5	89.4	77.8	88.3	77.3	81.9	87.5	80.9	86.4	80.0	85.5	14.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.2	85.4	79.4	152.5	84.6	78.4	147.6	83.9	45.8	87.8	44.7	86.2	43.7	86.0	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	23.3	21.9	42.2	94.3	1.8	2.6	40.9	96.1	39.8	97.7	38.8	99.1	37.6	101.0		
(5)			38.9	84.6	2.0	2.4	37.4	86.3	36.3	87.7	35.1	89.1	33.6	90.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.2	81.9	81.9	80.4	77.5	71.9	68.5	66.1	67.6	70.7	76.2	79.3	81.2		
	DBStd	6.54	2.48	2.41	2.11	2.53	4.15	4.36	4.24	3.88	3.48	2.51	2.16	2.68		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	90	0	0	0	0	6	21	40	20	3	0	0	0		
	CDD50	9206	989	893	941	825	679	555	499	547	622	811	879	966		
	CDD65	3820	524	473	476	375	219	126	74	102	175	346	429	501		
	CDH74	33285	5855	5216	4715	2772	1110	332	187	373	1062	2507	3871	5285		
	CDH80	9706	2166	1861	1484	637	107	2	6	9	95	498	1062	1779		
(14) Wind	WSAvg	11.9	12.4	12.2	13.4	13.2	11.6	12.0	11.3	11.4	10.2	11.5	11.9	11.7		
(15) Precipitation	PrecAvg	34.8	7.9	8.5	4.2	2.0	1.4	1.1	0.6	0.8	0.6	0.7	1.8	5.2		
	PrecMax	70.1	19.2	27.7	19.4	8.3	9.6	6.6	3.5	5.7	3.9	4.0	11.8	33.5		
	PrecMin	15.4	1.5	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6		
	PrecStd	15.0	5.2	7.8	4.6	2.5	1.9	1.5	0.8	1.4	0.9	0.9	2.4	6.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.7	93.0	90.2	87.9	84.3	80.3	80.1	81.1	86.0	87.3	89.2	91.6		
		MCWB	80.0	79.2	78.5	75.0	72.3	70.2	70.5	65.7	62.8	74.3	75.7	79.9		
	2%	DB	90.6	90.4	88.6	86.3	82.3	79.0	77.5	79.0	81.9	85.0	87.3	89.2		
		MCWB	79.1	78.2	77.0	74.8	71.4	69.3	67.8	66.7	69.7	74.0	76.3	78.2		
	5%	DB	89.2	89.1	87.6	85.2	81.1	77.6	76.2	77.5	80.8	83.9	86.3	88.0		
		MCWB	78.3	78.0	76.3	74.4	70.6	68.4	66.4	66.1	68.8	73.0	75.4	77.4		
	10%	DB	87.9	87.8	86.4	84.0	79.6	76.2	74.7	76.4	79.5	82.7	85.3	86.8		
		MCWB	77.3	77.7	75.7	73.8	69.7	67.4	65.1	65.7	68.1	72.0	74.7	76.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.5	82.5	81.7	78.8	76.8	75.4	75.3	73.8	75.1	78.8	82.4	83.4		
		MCDB	90.0	88.1	87.5	85.1	81.1	77.5	75.9	77.3	81.1	84.8	87.1	88.7		
	2%	WB	81.8	81.3	80.3	77.4	74.8	72.4	70.8	71.0	73.5	77.1	79.6	81.1		
		MCDB	87.9	87.0	85.8	83.1	79.3	76.3	75.0	75.6	78.9	82.6	84.9	86.6		
	5%	WB	80.7	80.4	79.0	76.6	73.0	70.9	68.8	69.3	71.9	75.5	77.9	79.9		
		MCDB	86.3	85.9	84.4	82.3	78.5	75.3	73.7	74.5	78.1	80.9	83.8	85.6		
	10%	WB	79.6	79.5	77.7	75.6	71.6	69.3	67.3	68.1	70.4	74.2	76.6	78.9		
		MCDB	85.3	85.1	83.4	81.7	77.7	74.2	72.9	73.6	77.0	80.1	82.8	84.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.1	12.2	12.8	15.3	15.0	17.9	17.5	18.2	14.0	12.3	11.4		
		MCDBR	12.7	13.3	13.8	14.3	15.8	15.0	17.8	18.7	18.2	14.7	13.1	11.8		
	5% WB	MCWBR	4.9	4.9	5.4	6.2	7.5	7.7	9.3	9.4	8.7	6.2	5.9	4.6		
		MCWBR	11.0	11.3	11.7	13.2	14.9	12.0	14.5	16.1	15.5	13.3	12.0	10.9		
(40) Clear-Sky Solar Irradiance	taub	0.405	0.399	0.386	0.364	0.347	0.334	0.321	0.330	0.359	0.385	0.390	0.401			
	taud	2.511	2.535	2.555	2.604	2.607	2.630	2.649	2.608	2.516	2.469	2.495	2.492			
	Ebn at Noon	298	297	292	287	281	279	287	294	296	297	301	299			
	Edh at Noon	36	35	33	29	27	25	26	29	34	37	37	37			
(44) All-Sky Solar Radiation	RadAvg	2032	1931	1778	1595	1375	1233	1387	1652	1991	2197	2234	2228			
	RadStd	176	209	166	143	91	115	94	101	116	124	251	213			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page